

Nov 10

Find $\int \frac{x - \arctan(x)}{x^3} dx$

$$x - \arctan(x) = x - \left(x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots \right)$$

$$= \frac{x^3}{3} - \frac{x^5}{5} + \frac{x^7}{7} - \dots$$

$$\frac{x - \arctan(x)}{x^3} = \left(\frac{x^3}{3} \cdot \frac{1}{x^3} \right) - \left(\frac{x^5}{5} \cdot \frac{1}{x^3} \right) + \left(\frac{x^7}{7} \cdot \frac{1}{x^3} \right) - \dots$$

$$= \frac{1}{3} - \frac{x^2}{5} + \frac{x^4}{7} - \dots$$

$$\int \frac{x - \arctan(x)}{x^3} dx = \int \frac{1}{3} - \frac{x^2}{5} + \frac{x^4}{7} - \dots$$

$$= \left(\frac{x}{3} - \frac{x^3}{3 \cdot 5} + \frac{x^5}{5 \cdot 7} - \dots \right)$$

↑
Denom of mult?

$$\arctan(x) = \int \frac{1}{1+x^2}$$

$$\arctan(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \quad |x| < 1$$

$$C + x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

$0 = \arctan(0) = C \Rightarrow C = 0$

$$\text{So } \arctan(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$$

for $|x| < 1$

$$\arctan(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

It's
easier
to expand +
treat as
a polynomial?